TASK 1: YOUNGEST CATEGORY
Kits (Ages 6–8) | Difficulty: Easy/Medium | Answer type: Multiple choice

Buzzy's Fruit School

AI Concept: Supervised Learning	CT Skills: Pattern Recognition, Abstraction	Time: ~3 minutes	Interactive: No

Story

Buzzy is a little robot bee who works in a fruit garden. Buzzy has never seen fruit before, so the gardener needs to teach Buzzy which fruits are SWEET and which fruits are SOUR.

The gardener picks up fruits one at a time and tells Buzzy what each one tastes like:

Fruit	Colour	Size	Texture	Gardener says:
🍓 Strawberry	Red	Small	Rough	SWEET
🍋 Lemon	Yellow	Small	Smooth	SOUR
🍒 Cherry	Red	Small	Smooth	SWEET
🍊 Grapefruit	Yellow	Big	Smooth	SOUR

Buzzy looks at the table very carefully. Buzzy notices something about the colours!

Now, a new fruit arrives: a **Red Apple**. It is red, big, and smooth.

Question: What will Buzzy say about the red apple?

A) "SWEET!" — because all the red fruits were sweet

B) "SOUR!" — because all the smooth fruits were sour

C) "I don't know" — because Buzzy has never seen an apple before

D) "SWEET!" — because the apple is big

Answer

Correct Answer: A)

"SWEET!" — because all the red fruits were sweet.

Explanation

If we look at the gardener's teaching table, we can spot a clear pattern based on colour:

The two red fruits (strawberry and cherry) were both labelled SWEET. The two yellow fruits (lemon and grapefruit) were both labelled SOUR. From these examples, Buzzy can learn a simple rule: Red = Sweet, Yellow = Sour.

When the red apple arrives, Buzzy applies this learned rule and predicts SWEET — and the reasoning (because all the red fruits were sweet) matches the pattern Buzzy found in the training examples.

Why the other answers are wrong:

B) is incorrect because not all smooth fruits were sour. The cherry is smooth AND sweet. So "smooth = sour" is not a consistent rule from the examples.

C) is incorrect because the whole point is that Buzzy learns rules from examples and applies them to new items it has never seen. Buzzy doesn't need to have seen an apple before — it uses the pattern it learned.

D) has the right prediction (SWEET) but the wrong reasoning. Size doesn't consistently predict taste in the training data: both small strawberry and small lemon differ in taste, and the big grapefruit is sour.

Connection to Informatics: Supervised Learning

This task introduces the foundational concept of supervised learning, which is how many AI systems learn. In supervised learning, a computer program (like Buzzy) is given a set of labelled examples — called training data — where each example comes with the correct answer. The program looks for patterns (or "rules") in the features of the training data that predict the labels.

Once the program has learned a rule, it can use that rule to make predictions about new, unseen data. This is exactly how many real-world AI systems work: email spam filters learn from thousands of emails that humans have labelled as "spam" or "not spam," and voice assistants learn to recognise spoken words from millions of labelled audio recordings.

The key insight for young learners is that AI doesn't "know" things on its own — it learns from examples that people give it, finds patterns, and then applies those patterns to new situations. If the examples have a clear pattern, the AI can make good predictions even for things it has never seen before.